AWS-EC2 Instances

- **What are AWS Regions?**
- In **Amazon Web Services (AWS)**, a **Region** is a **geographically isolated area** where AWS has **multiple data centers** (called **Availability Zones**) to host and run cloud infrastructure and services.
- Each AWS Region is completely **independent** and designed to offer **low latency**, **fault tolerance**, and **high availability**.
- **What are AWS Availability Zones (AZs)?**
- An **Availability Zone (AZ)** in AWS is one or more **physically separate data centers** within a single **AWS Region**, designed for **high availability**, **fault isolation**, and **low-latency connectivity**.
- Each AZ has its own **power, cooling, and networking**, but is **connected to other AZs in the same region with high-speed, low-latency links**.
- ✓ **Region vs. Availability Zone: Key Differences**

Feature	Regions	Availability Zones
Definition	A geographical area containing multiple AZs.	A single data center within a Region.
Number per AWS Region	1 Region per geographic location.	2 or more AZs per Region.
Scope	Contains multiple AZs.	Contains multiple physical data centers.
Fault Tolerance	A failure in one Region does not impact others.	AZs are independent but can back up each other.
Use Case	Choose a Region based on latency, compliance, and cost.	Use multiple AZs for high availability and failover.

AWS Multi-Region & Multi-AZ Best Practices

- Multi-AZ Deployment – Use multiple AZs in a Region for fault tolerance.
- Multi-Region Deployment – Deploy resources across Regions for disaster recovery.
- Elastic Load Balancing (ELB) – Distribute traffic across AZs.
- Amazon RDS Multi-AZ Replication – Automatic failover for databases.
- Amazon S3 Cross-Region Replication – Backup data in another Region

Introduction to EC2 service

- Amazon Elastic Compute Cloud (Amazon EC2) provides on-demand, scalable computing capacity in the Amazon Web Services (AWS) Cloud. Using Amazon EC2 reduces hardware costs so you can develop and deploy applications faster.

Features of Amazon EC2

- **Amazon EC2 (Elastic Compute Cloud)** is a core AWS service that provides **resizable virtual servers (instances)** in the cloud.
- It lets users run applications on virtual machines without managing the physical hardware.
- Each instance type offers a different balance of compute, memory,network, and storage resources.

❖ What is an Amazon Machine Image (AMI)?

- An **Amazon Machine Image (AMI)** is a **pre-configured template** used to **launch EC2 instances** in AWS.
  - ✓ It contains all the information required to boot up a virtual server, including:
  1. The **operating system**
  2. Pre-installed **software/packages**
  3. **Application code**
  4. **Configuration settings**
  5. Optional: Custom scripts, data, etc.

❖ Instance types

- **EC2 instance types** define the **hardware configuration** of virtual machines (EC2 instances) in AWS.
- Each instance type is designed for specific use cases like general computing, memory-intensive tasks,
- ✓ storage-heavy workloads, or GPU processing.

1. General Purpose - t2, t3 (cpu, mem balanced)
2. Compute Optimize - c2, c4 (cpu focused, mem)
3. Memory Optimized - m4 (cpu, mem focused)
4. Storage Optimize - (cpu mem balanced, SSD focused)
5. Graphics Optimize - (Graphics card)

- **Status Check**

1. 2/2 - Everything is fine
2. 1/2 - Software / OS / Setting / booting - troubleshoot/check logs /var
3. 0/2 - hardware - AWS Team (Ticket Rise) - instance-ID

❖ Amazon EBS volumes

(Amazon Elastic Block Store) These volumes behave like **hard drives**, allowing you to store **data persistently**

❖ Instance Store Volumes

- **Instance Store Volumes** (also called **ephemeral storage**) are **temporary block-level storage** physically attached to the host machine running the EC2 instance.
- They offer **high-speed storage** but **do not persist** after the instance is stopped, terminated, or fails.

❖ Placement Strategies:

- There are **three types of placement strategies**:

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1. Cluster Placement Group

- **Purpose**: Low-latency, high-throughput network performance.
- **How it works**: Instances are placed physically close together in a single Availability Zone.
- **Use cases**: High-performance computing (HPC), big data workloads, distributed databases.
- **Pros**: Fast communication between instances (up to 100 Gbps).
- **Cons**: If capacity is not available, instance launches may fail.

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2. Spread Placement Group

- **Purpose**: Maximize availability by placing instances on distinct hardware.
- **How it works**: Each instance is placed on a separate rack (power, network, etc.).
- **Use cases**: Critical applications that require isolation to reduce simultaneous failures.
- **Pros**: Reduces the risk of simultaneous hardware failures.
- **Cons**: Limited to 7 running instances per Availability Zone per spread group.

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3. Partition Placement Group

- **Purpose**: Control how instances are distributed across partitions (groups of racks).
- **How it works**: Instances are divided into partitions, each isolated from the others.
- **Use cases**: Large distributed and replicated workloads like Hadoop, Cassandra, etc.
- **Pros**: Isolation + ability to use many instances.
- **Cons**: More complex to manage than other strategies.

- **Summary Table**

Placement Type	**AZ Scope**	**Isolation Level**
Cluster	Single AZ	None
Spread	Multi-AZ	Host-level
Partition	Multi-AZ	Rack-level

❖ How does AWS EC2 handle instance failures?

➢ If an EC2 instance fails, AWS provides multiple fault-tolerant mechanisms:

1. **Auto Recovery:**
   - AWS automatically detects failure and moves the instance to a new host.
   - Requires enabling CloudWatch Recovery Actions.

2. **EC2 Auto Scaling:**
   - Detects failures and launches replacement instances.

3. Elastic Load Balancer (ELB):
   - Distributes traffic away from unhealthy instances.

4. Use Multi-AZ Deployments:
   - Deploy instances across multiple Availability Zones (AZs) for high Availability

❖ How can you migrate an EC2 instance to another AWS Region?

➢ ***To migrate an EC2 instance across Regions***:

- **Create an AMI (**Amazon Machine Image)

1. aws ec2 create-image
2. Instance-id i-xxxxxx
3. Name "my-instance-image"

➢ **Copy AMI to another Region**

1. aws ec2 copy-image
2. source-region us-east-1
3. source-image-id ami-xxxxxx
4. Destination-region ap-south-1
5. Name "copied-instance"

➢ Launch a new EC2 instance in the target

❖ What is a Security Group in AWS?

1. A Security Group in AWS acts as a virtual firewall that controls inbound and outbound traffic to EC2 instances and other supported resources. It defines which traffic is allowed to reach or leave the associated resource, based on rules configured by the user.
2. Security Groups are associated with Elastic Network Interfaces (ENIs), and each EC2 instance must be associated with at least one Security Group.

❖ How Security Groups Work

Key Characteristics:

- Stateful: If you allow an inbound request, the response traffic is automatically allowed, regardless of outbound rules.
- Default Deny-All: By default, no inbound traffic is allowed; outbound traffic is allowed by default.
- Attached to ENIs: Security Groups are applied to network interfaces, not to subnet

❖ Common Ports for Remote Access

Protocol	**Port**	**Used For**
TCP	22	SSH (Linux remote access
TCP	3389	RDP (Windows remote access)
TCP	80	HTTP (Web traffic)

TCP	443	HTTPS (Secure web traffic)
TCP	3306	MySQL Database
TCP	5432	PostgreSQL Database

❖ **How do you configure an EC2 instance for high availability?**

Answer:

To ensure high availability of an EC2 instance:

1. Deploy instances across multiple AZs (Availability Zones) to prevent single points of failure.
2. Use an Elastic Load Balancer (ELB) to distribute traffic.
3. Enable Auto Scaling to automatically replace failed instances.
4. Use Elastic IPs to maintain consistent public access in case of failures.
5. Implement regular backups using AMIs and snapshots

❖ **Explain the difference between Security Groups and Network ACLs.**

Feature	Security Groups	Network ACLs
Level	Instance level	Subnet level
Stateful/Stateless	**Stateful** (automatically allows response traffic)	**Stateless** (rules must allow both inbound & outbound separately)
Default Behavior	Denies all **inbound**, allows all **outbound**	Allows all **inbound** and **outbound** by default
Rules Evaluation	Evaluates **only allowed** rules	Evaluates **both allow** and deny rules

Use Case:

- Security Groups are used to control access to EC2 instances.
- Network ACLs provide an additional layer of security at the subnet level

❖ **What is the difference between a Public and Private EC2 instance?**

1. **Public EC2 Instance:** Has a public IP and is directly accessible over the internet.
2. **Private EC2 Instance:** Only has a private IP and requires a NAT Gateway, Bastion Host, or VPN to access the internet.

◈ **Best Practice**: Keep sensitive workloads in a private subnet and use a bastion host or VPN for secure access.